

Type-Aware Failure Prediction for Chart Reasoning in VLMs

Mid-term status, for TA guidance

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<https://github.com/the-neemon/type-aware-vlm-failures>

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The question

Two ways to answer a chart question wrongly

Structural failure (misread)

- Wrong bar, misjudged tick, miscounted slices
- The answer *is* in the image
- Repair: look again (crop, re-ask)

Fabrication

- States a value or label the chart does not contain
- Looking again cannot help
- Repair: abstain

Today's reliability tools collapse both into one "do not trust" score.

Type Separability Hypothesis: the model's internal state *before* it answers is linearly different for a coming misread and a coming fabrication. A collapse of the two into one signal is a reportable outcome, not a failure.

Experiments at a glance

	What it asks	Status
E1	Can a probe predict <i>whether</i> the answer will be wrong?	done
E2	Is the probe just reading surface features of the input?	mostly done
E3	Are the misread and fabrication signals different (cross-transfer)?	done
E4	Does the failure type decide which repair works?	not yet run
E5	A controller that picks the repair from the probes	not yet run

Two data arms: **ChartQA** (real charts, naturally occurring errors) and a **synthetic bar-chart arm** (errors induced on purpose).

Pipeline

Models (frozen, fp16, greedy)

- Qwen2.5-VL-7B (28 layers, $d = 3584$)
- LLaVA-NeXT-7B, Mistral LM (32 layers, $d = 4096$)
- ChartGemma dropped

Data

- ChartQA test: 2,500 questions (1,250 human, 1,250 augmented)
- ChartQA val_human: 960 (Qwen)
- Two synthetic pilots: 2,400 questions

Activations

- Forward hooks inside the *same* generate() call that produced the labelled answer
- Every layer, 4 pooled positions: image-token mean/max, last prompt token, prompt mean

Probes

- L_2 logistic regression on one vector, scored by AUROC
- Splits by *chart*, never by question
- Figure-level bootstrap CIs
- Always next to a baseline that sees no activations

Naturalistic arm: ChartQA

Model	Overall	Human	Augmented
Qwen2.5-VL-7B	87.3	79.8	94.8
LLaVA-NeXT-7B	52.9	44.6	61.1

Relaxed accuracy (%), numeric within 5% of gold.
Qwen is wrong on **317** questions, LLaVA on **1,178**.

What the errors actually are

Label	Qwen		LLaVA	
	<i>n</i>	%	<i>n</i>	%
STRUCTURAL	71	22	583	49
COMPUTATION	93	29	435	37
NOT_AN_ERROR	112	35	69	6
AMBIGUOUS	40	13	74	6
FABRICATION	1	0.3	17	1.4
errors	317		1,178	

Same LLM annotator (Claude Opus 5.5) for both models: saw only image, question, gold and answer. Blind re-label of 50 Qwen errors: $\kappa = 0.917$. Human agreement check not yet run.

- **Fabrication is essentially absent** for both models (1 and 17), even though LLaVA errs 4× as often: questions are written about content the chart contains
- **35% of Qwen's "errors" are not errors**, but only 6% of LLaVA's: a strong model's leftover errors are mostly scoring artefacts
- LLaVA's largest class is STRUCTURAL: garbled printed values (486 for 4,863), adjacent bars, gridline snapping
- COMPUTATION is large for both; the proposal's two-class taxonomy had no slot for it

⇒ ChartQA cannot supply fabrication examples; this is why the synthetic arm exists.

Label noise, and where each model's errors come from

	Qwen		LLaVA	
	Human	Augm.	Human	Augm.
Questions	1,250	1,250	1,250	1,250
Scored wrong	252	65	692	486
Real errors	140	25	607	428
STRUCTURAL	52	19	192	391
COMPUTATION	87	6	409	26
Gold answer wrong	16	18	22	22

- Gold answers wrong: 34 (Qwen) and 44 (LLaVA); on Qwen's augmented questions that is 28% of the "errors"
- Counting NOT_AN_ERROR as correct: Qwen 87.3% $\rightarrow$ 91.8%, LLaVA 52.9% $\rightarrow$ 55.6%
- LLaVA's error type follows the question source: arithmetic on human-written questions, misreads on augmented ones

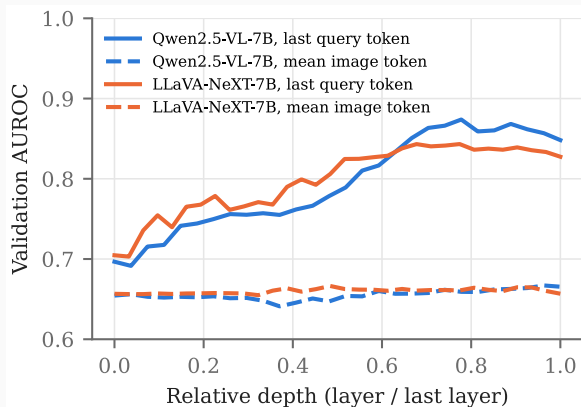
Every probe trained on relaxed-accuracy labels inherits this noise.

E1: predicting whether the answer will be wrong

Test AUROC	Probe [95% CI]	5 splits	Resid.	Surface
Qwen, all errors	0.877 [.83, .92]	.862 $\pm$.02	.803	.715
Qwen, without NOT_AN_ERROR	0.932 [.90, .96]	.910 $\pm$.02	.901	.786
LLaVA, all errors	0.798 [.76, .84]	.801 $\pm$.02	.783	.608
LLaVA, without NOT_AN_ERROR	0.819 [.78, .85]	.810 $\pm$.01	.798	.636

- Surface: question source, length, image tokens, numeric gold. Resid.: those features regressed out of the activations; most of the signal survives
- 5 splits: mean $\pm$ s.d. over five figure splits; Qwen's reported split is above the mean
- Removing label noise *raises* the probe; within human-written questions only, 0.895 (Qwen) and 0.850 (LLaVA)
- Unseen charts (960 Qwen validation questions): 0.772, against 0.829 on the human-written test questions; probabilities calibrated (ECE 0.03 Qwen, 0.05 LLaVA)

E1: where the signal lives



- Last prompt token, building with depth, plateau in the last third (Qwen best layer 21/28, LLaVA 24/32)
- Image tokens flat at 0.62–0.68, near the surface baseline
- Differs from HALP (Qwen best on visual features for photo hallucination): for charts, the state that has read the question matters

E3 on real charts: misreads vs. arithmetic errors

AUROC (5-fold CV by chart)	Qwen	LLaVA
Misread vs. correct	0.892 [.85, .93]	0.813 [.79, .83]
surface baseline	0.651	0.536
retrieval questions only	0.913	0.805
Computation vs. correct	0.904 [.87, .93]	0.949 [.94, .96]
surface baseline	0.818	0.821
arithmetic questions only	0.845	0.836
Misread vs. computation	0.897 [.84, .94]	0.961 [.95, .97]
surface baseline	0.693	0.845
Cross: misread probe on comp. vs. correct	0.820	0.586

- Qwen 71 misreads / 93 computation errors; LLaVA 583 / 435; too few fabrications for a probe (1 and 17)
- LLaVA's probes are type-specific; Qwen's misread probe also fires on arithmetic errors (surface features alone: 0.765)
- Question wording gives part of the arithmetic signal away (layer 0: 0.865 for LLaVA), but probes trained *within* one question type still work
- Surface features regressed out (Qwen): all three probes stay at 0.83 to 0.92

Synthetic arm

Making fabrication happen on purpose

Ask about a category that is *not* on the chart.

- **Pilot 1:** 300 bar charts. Same wording twice per chart: “What is the value of Mango?” (shown) vs. “... of Guava?” (missing)
- **Pilot 2:** 300 harder charts, 1,800 questions
 - Lookalike pairs (Dubai/Dublin, Eagle/Falcon): chart shows at most one
 - Three families: *value*, *compare*, *neighbor* (“Which bar is right of Guava?”)
 - Values never multiples of 5, axes up to 200, sparse ticks

Two guards against leakage

- Every contrast is *within one question template* (the template alone predicts the intended type at AUROC 1.000)
- Outcomes scored from the actual answer, never from what the question was designed to cause

How the models answered

		Qwen	LLaVA
Shown, right	value	554/600	185/600
	compare	300/300	209/300
	neighbor	294/300	255/300
Missing, value	rejected	15	0
	“0”	494	232
	fabricated	91	368
Missing, compare	fabricated	37	101
Missing, neighbor	fabricated	300	290

- **Neither model refuses:** 15/1,200 (Qwen), 0 (LLaVA)
- Qwen mostly says “0”; when it invents, it reads the lookalike bar (asked Dubai, reports Dublin)
- LLaVA snaps to gridlines (52 → 50): only 31% of shown values right
- “0” is treated as a refusal expressed as a number

Value template gives four classes: C correct, S misread, F invented number, Z “0”.

Do the models know the bar is missing? Yes.

Test AUROC	Qwen	LLaVA
value	0.999	0.977
compare	0.991	0.959
neighbor	1.000	0.996
lookalikes only	0.98–1.00	0.92–0.99
text-only baseline	0.49–0.52	0.49–0.52
image-token probe	0.500	0.500

- Both models **represent that the name is missing, then answer anyway**
- Text-only baseline at chance: every name is shown on some charts
- Image-token control is exactly 0.500 by construction (image precedes the question, attention is causal)
- Early in Qwen (layer 2), later in LLaVA (~9)

E3 on synthetic charts: cross-transfer matrix

Probe (trained on)	Qwen2.5-VL-7B			LLaVA-NeXT-7B		
	S vs C	F vs C	F vs Z	S vs C	F vs C	F vs Z
structural (S vs C)	0.887	0.594	0.400	0.776	0.879	0.215
metadata baseline	0.814	0.557	0.543	0.512	0.522	0.537
fabrication (F vs C)	0.576	1.000	0.286	0.680	0.988	0.243
invented vs. zero (F vs Z)	0.336	0.138	0.984	0.520	0.226	0.986

- **Qwen: largely type-specific.** Each probe strong on its own task, weak on the other
- **LLaVA: misread probe is not specific.** It fires *more* on fabrications than on misreads, a general “not grounded” direction
- **Both:** the “fabrication” probe is really a missing-bar detector (scores “0” as high as invented numbers)

5-fold CV by chart, out-of-fold AUROC, synthetic value questions.

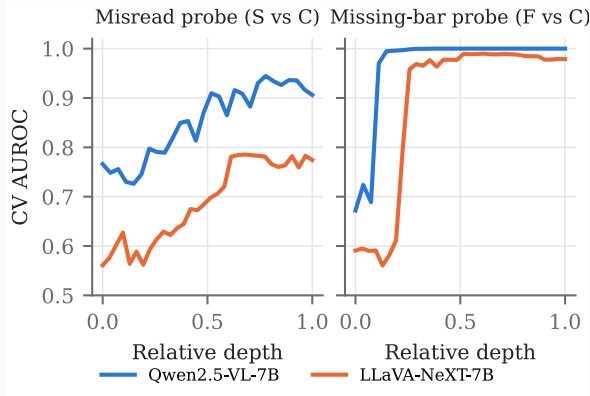
Synthetic misreads: a chart property, and no transfer

Gridline position predicts synthetic misreads (misread vs. correct, AUROC)

	Qwen	LLaVA
Activation probe	0.887	0.776
Metadata + gridline distance (declared in advance)	0.853	0.642
Metadata + distance relative to the value (post hoc)	0.920	0.860

- Scoring tolerates a 5% *relative* error: rounding 34 to 30 is wrong, 134 to 130 is not
- **Synthetic probes on real charts:** misread probe on real misreads 0.597 (Qwen), 0.557 (LLaVA); missing-bar probe on LLaVA's 17 real fabrications 0.650, but equally on real misreads (0.669)
- So the synthetic misread signal may describe the chart, and the synthetic probes do not carry over to ChartQA

E3: where the type signals live



- Missing-bar signal saturates early (Qwen by layer 3, LLaVA ~8)
- Misread signal builds gradually, peaks in the last third
- Invented-vs-“0” only in the last layers: reads out the answer about to be produced, not an early warning

Where we stand

Verdict so far

Hypothesis: partly supported.

- Real charts: misreads and arithmetic errors separable in both models (0.897 Qwen, 0.961 LLaVA), also within one question type; type-specific in LLaVA
- Synthetic: both models encode that the bar is missing; in Qwen the misread and missing-bar probes are largely blind to each other's type
- Against: LLaVA's synthetic misread probe fires more on fabrications, and gridline position predicts synthetic misreads as well as either probe

Caveats

- On synthetic data, “fabrication” and “missing category” coincide, and the synthetic probes do not transfer to real charts
- Qwen has only 46 synthetic and 71 real misreads
- Labels are from one LLM annotator (blind self-agreement $\kappa = 0.917$); no two-person human check yet

Done

- Full-depth activation caches: both models, ChartQA test, both synthetic pilots; ChartQA validation (Qwen)
- Error labels for both models (317 Qwen, 1,178 LLaVA)
- E1 for both models, with five-split robustness, calibration and an unseen-chart test (Qwen)
- E2: surface baselines everywhere; residualised probes (binary both models, type probes Qwen)
- E3 for both models: synthetic matrix, ChartQA misread vs. arithmetic (also within question type), synthetic → ChartQA transfer

Not started

- Node-link diagrams, the “diagram” half of the title: the generator is written but has not been run

Timeline for the rest of the project

Final submission: 31 October. E4 code is written and tested.

- **3–10 Oct.** E4 cross-intervention matrix on every synthetic value-question error of both models (Qwen 631, LLaVA 1,015): crop, upsample, verify prompt, sampled re-ask; pre-registered difference-in-differences test. In parallel: two-person human κ on 200 errors (both models); LLaVA residualised type probes
- **11–20 Oct.** E5 controller: abstain when the missing-bar probe fires, re-look when the binary or misread probe fires; risk-coverage and AURC against binary selective prediction and the oracle. In parallel: out-of-range synthetic pilot, so fabrication no longer equals a missing category
- **21–31 Oct.** Final paper, code release, viva slides

Where we would like guidance

1. **Fabrication definition.** Is “asked about a missing category” an acceptable operationalisation, or should we add out-of-range quantities so fabrication is not identical to absence?
2. **Taxonomy.** Should COMPUTATION become a third type in the hypothesis, given it is the largest real error class and probes separate it from misreads?
3. **Annotation.** Is an LLM annotator acceptable if it passes a two-person human agreement check ($\kappa > 0.6$ on 200 items)?
4. **Label noise.** Should the NOT_AN_ERROR items (Qwen 112, LLaVA 69) be re-scored as correct in every experiment, or reported both ways?
5. **Scope of E4/E5.** Is running the repairs on synthetic charts only (where bar geometry is known) enough, or do we need ChartQA repairs too?

Thank you